Dynamic Causal Factor Model (DCFM) for Institutional Governance

A Graph-Based Ontological Framework for "Do No Harm" Decision-Making

Version 1.0
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Executive Summary

This document presents a Dynamic Causal Factor Model (DCFM) designed to evaluate institutional decisions through a "Do No Harm" lens, incorporating temporal analysis, interstitial impacts, and both human and machine-driven decision pathways. The framework uses a graph-based ontological structure that can be computationally rendered and analyzed, while remaining human-interpretable for governance bodies.

Core Innovation: This framework moves beyond traditional impact assessment by modeling the recursive, emergent, and interstitial effects of decisions across multiple timescales and domains, specifically designed to prevent the "Moloch trap" of rivalrous competition and extractive efficiency.

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1. Foundational Ontology

1.1 Core Entities

The DCFM is built upon a fabric of interconnected ontologies that represent the key domains of institutional impact:

Primary Entity Classes

```
Entity Types:
├── Actors
│   ├── Human Actors
│   │   ├── Individuals
│   │   ├── Communities
│   │   └── Collectives
│   ├── Institutional Actors
│   │   ├── Corporations
│   │   ├── Governments
│   │   ├── NGOs
│   │   └── Cooperatives
│   └── Non-Human Actors
│       ├── Algorithmic Systems
│       ├── AI Agents
│       └── Automated Processes
│
├── Resources
│   ├── Natural Capital
│   │   ├── Biotic Resources (ecosystems, species, genetic diversity)
│   │   ├── Abiotic Resources (minerals, water, atmosphere)
│   │   └── Regenerative Capacity (soil health, carbon sequestration)
│   ├── Human Capital
│   │   ├── Knowledge Systems (indigenous, scientific, tacit)
│   │   ├── Cognitive Agency (attention, decision-making capacity)
│   │   └── Social Capital (trust, cooperation, cultural practices)
│   ├── Manufactured Capital
│   │   ├── Infrastructure (physical, digital, institutional)
│   │   ├── Technology Systems
│   │   └── Financial Instruments
│   └── Commons
│       ├── Global Commons (atmosphere, oceans, biodiversity)
│       ├── Digital Commons (open-source, knowledge repositories)
│       └── Local Commons (community resources, seed banks)
│
├── Decisions
│   ├── Planned Actions
│   │   ├── Policy Interventions
│   │   ├── Investment Choices
│   │   └── Operational Changes
│   ├── Decision Mechanisms
│   │   ├── Human Deliberative
│   │   ├── Algorithmic Optimization
│   │   └── Hybrid Governance
```

```
│   └── Decision Context
│       ├── Constraints (resource, regulatory, technical)
│       ├── Incentive Structures
│       └── Power Dynamics
│
└── Impacts
    ├── Direct Impacts (first-order effects)
    ├── Interstitial Impacts (emergent, unplanned)
    ├── Recursive Impacts (self-modifying)
    └── Systemic Impacts (threshold effects, tipping points)
```

1.2 Core Relationships

The graph structure connects entities through typed relationships:

```
Relationship Types:
├── Causal Relationships
│   ├── CAUSES → (direct causation)
│   ├── ENABLES → (creates conditions for)
│   ├── INHIBITS → (prevents or reduces)
│   ├── AMPLIFIES → (increases magnitude of)
│   └── DAMPENS → (reduces magnitude of)
│
├── Temporal Relationships
│   ├── PRECEDES → (temporal ordering)
│   ├── CONCURRENT_WITH → (simultaneous)
│   ├── DELAYED_BY → (time lag specification)
│   └── CASCADES_TO → (sequential triggering)
│
├── Resource Relationships
│   ├── EXTRACTS_FROM → (resource depletion)
│   ├── REGENERATES → (resource restoration)
│   ├── TRANSFORMS → (resource conversion)
│   └── ACCUMULATES_IN → (resource concentration)
│
├── Power Relationships
│   ├── GOVERNS → (authority over)
│   ├── INFLUENCES → (shapes behavior of)
│   ├── DEPENDS_ON → (requires for operation)
│   └── COMPETES_WITH → (rivalrous dynamic)
│
└── Boundary Relationships
    ├── WITHIN_DOUGHNUT → (social foundation met, ecological ceiling respected)
    ├── BELOW_FOUNDATION → (social need unmet)
    ├── ABOVE_CEILING → (planetary boundary breached)
    └── AT_THRESHOLD → (approaching critical limit)
```

1.3 Ontological Principles

Principle 1: Interconnection
Every entity exists in a web of relationships. No decision can be evaluated in isolation.

Principle 2: Emergence
The behavior of the system is not reducible to the sum of its parts. Interstitial effects are properties of the network structure.

Principle 3: Temporal Multiplicity
Impacts unfold across multiple timescales simultaneously. A decision may benefit the present while harming the

deep future.

Principle 4: Recursive Causation
Decisions change the decision-maker. Using AI to optimize decisions reduces human cognitive capacity over time.

Principle 5: Boundary Respect
Planetary boundaries and social foundations are non-negotiable constraints, not externalities to be optimized against.

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2. The Causal Factor Model Architecture

2.1 Three Layers of Impact Analysis

The DCFM evaluates every institutional decision across three nested layers:

Layer 1: Direct Impacts (Planned Outcomes)

Definition: The immediate, intended consequences of a decision as designed by the decision-maker.

Evaluation Questions:

- What is the explicit goal of this decision?
- What resources are required for implementation?
- Who are the direct beneficiaries?
- What metrics define "success" from the institutional perspective?

Graph Representation:

```
Decision Node → INTENDED_OUTCOME → Direct Impact Node
          → REQUIRES → Resource Node
          → BENEFITS → Actor Node
```

Example:
Decision: Launch a new AI hiring tool
Direct Impact: Reduce time-to-hire by 40%
Resources Required: Training data (past hiring decisions), computational infrastructure
Beneficiaries: HR department efficiency

Layer 2: Interstitial Impacts (The Gaps and Emergent Effects)

Definition: The unintended consequences and feedback loops that occur between the planned action and the system's response. These are the "gaps" where complexity lives—the space between intention and outcome.

Evaluation Questions:

- What systemic feedback loops does this decision create or amplify?
- Who are the indirect stakeholders affected by this decision?
- What power dynamics shift as a result?
- What dependencies are created or broken?
- What knowledge or agency is transferred, concentrated, or eroded?

Graph Representation:

```
Direct Impact Node → TRIGGERS → Feedback Loop Node
           → CREATES_DEPENDENCY → New Dependency
           → SHIFTS_POWER → Power Redistribution Node
           → ERODES → Agency/Knowledge Node
```

Example (continued):
Interstitial Impacts of AI Hiring Tool:

- Algorithmic bias perpetuates historical discrimination (AMPLIFIES existing inequity)
- Human HR staff lose skill development in candidate assessment (ERODES cognitive agency)
- Vendor lock-in creates dependency on proprietary system (CREATES_DEPENDENCY)
- Candidates' data harvested without meaningful consent (EXTRACTS_FROM digital commons)
- Homogenization of workforce culture as algorithm optimizes for pattern-matching (REDUCES diversity)

Layer 3: Recursive Impacts (Self-Modifying Effects)

Definition: How the decision fundamentally changes the nature of the decision-maker and the decision-making process itself. These are meta-level impacts on institutional and human capacity.

Evaluation Questions:

- How does this decision change our future decision-making capacity?
- Does this enhance or reduce our collective wisdom?
- What forms of knowledge become invisible or devalued?

- How does this affect our relationship to uncertainty and complexity?
- Does this make us more or less resilient to future shocks?

Graph Representation:

```
Decision Node → MODIFIES → Institutional Capacity Node
        → CONSTRAINS → Future Decision Space
        → SHAPES → Cognitive Infrastructure
        → AFFECTS → Collective Wisdom
```

Example (continued):
Recursive Impacts of AI Hiring Tool:

- Organization loses ability to make hiring decisions without algorithmic assistance (REDUCES institutional capacity)
- Intuition and relational skills in hiring atrophy (ERODES tacit knowledge)
- Future decisions increasingly delegated to "black box" systems (CONSTRAINS decision space)
- Organization becomes less able to recognize and value non-quantifiable human qualities (SHAPES values)

2.2 The Moloch Trap Detector

A critical component of the DCFM is the **Moloch Trap Detector**—a pattern recognition system that identifies when decisions are driven by rivalrous competition rather than actual value creation.

Moloch Patterns:

1. **The Race to the Bottom**
   - Condition: If competitor does X, we must do X or lose market position
   - Result: All actors forced into harmful behavior
   - Detection: COMPETES_WITH relationship + negative collective outcome
2. **The Efficiency Paradox**
   - Condition: Optimization for single metric degrades system health
   - Result: Short-term gains, long-term fragility
   - Detection: EXTRACTS_FROM resources + REDUCES redundancy
3. **The Externality Export**

- Condition: Costs pushed onto commons, future generations, or weaker actors
- Result: Private profit, socialized harm
- Detection: BENEFITS actor + HARMS commons + temporal mismatch

4. **The Addiction Loop**
   - Condition: Solution creates dependency requiring more of same solution
   - Result: Escalating commitment to failing strategy
   - Detection: CREATES_DEPENDENCY + REQUIRES repeated intervention + DEGRADES alternatives

Moloch Trap Response Protocol: When a Moloch pattern is detected, the decision must be:

1. Flagged for mandatory Wisdom Council review
2. Evaluated for collective action alternatives
3. Assessed for regulatory intervention necessity
4. Redesigned to eliminate competitive pressure as driver

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3. Temporal Analysis Framework

3.1 Three Timescales of Evaluation

Every decision must be evaluated across three temporal horizons:

Timescale 1: Intergenerational Past (The Debt)

Time Horizon: Historical to present (focusing on 50-500 years)

Core Questions:

- Does this decision perpetuate or heal historical harms?
- Does it restore or further exploit communities and ecosystems harmed by past extraction?
- Does it acknowledge and address accumulated "ecological debt" and "social debt"?

Key Concepts:

- **Biopiracy Audit:** Does this decision rely on appropriated indigenous knowledge or genetic resources without consent and benefit-sharing?

- **Colonial Continuity:** Does this replicate extractive patterns established during colonial periods?
- **Intergenerational Justice:** Does this honor obligations to past generations' sacrifices and future generations' needs?

Evaluation Metrics:

- Reparative vs. Extractive Score
- Indigenous Sovereignty Respect Index
- Historical Pattern Disruption Measure

Graph Relationships:

```
Decision Node → ACKNOWLEDGES → Historical Harm Node
      → REPAIRS → Damaged Relationship
      → PERPETUATES → Colonial Pattern
      → HONORS → Intergenerational Obligation
```

Timescale 2: The Present Doughnut (The Now)

Time Horizon: 0-10 years

Core Questions:

- Does this decision meet human needs while respecting planetary boundaries?
- Does it move us toward or away from the "safe and just space"?
- Does it reduce or increase inequality?

Key Framework: Raworth's Doughnut

Social Foundation (Inner Boundary):

- Food security
- Water and sanitation
- Health
- Education
- Income and work
- Peace and justice
- Political voice

- Social equity
- Gender equality
- Housing
- Networks
- Energy

Ecological Ceiling (Outer Boundary):

- Climate change
- Ocean acidification
- Chemical pollution
- Nitrogen and phosphorus loading
- Freshwater withdrawals
- Land conversion
- Biodiversity loss
- Air pollution
- Ozone layer depletion

Evaluation Protocol:

```
For each dimension of social foundation:
  IF decision IMPROVES access THEN +1
  IF decision HARMS access THEN -1
  IF decision NEUTRAL THEN 0

For each planetary boundary:
  IF decision REDUCES pressure THEN +1
  IF decision INCREASES pressure THEN -1
  IF decision NEUTRAL THEN 0

Doughnut Score = (Social Foundation Score) - (Ecological Ceiling Score)
Target: Maximize Social Foundation while minimizing Ecological Ceiling breach
```

Graph Relationships:

```
Decision Node → MOVES_TOWARD / MOVES_AWAY → Doughnut Center
          → IMPROVES / DEGRADES → Social Foundation Dimension
          → RESPECTS / BREACHES → Planetary Boundary
          → REDISTRIBUTES / CONCENTRATES → Resource Access
```

Timescale 3: Deep Future (Existential Horizon)

Time Horizon: 10-100+ years

Core Questions:

- Does this decision create irreversible changes?
- Does it cross critical thresholds or tipping points?
- Does it enhance or reduce civilizational resilience?
- Does it create "lock-in" effects that constrain future adaptation?

Key Concepts from Schmachtenberger:

- **Self-Terminating Technologies:** Does this create risks that could end the human project?
- **Complexity Debt:** Does this create problems that are harder to solve than the original problem?
- **Option Space:** Does this expand or contract the range of possible futures?

Existential Risk Categories:

1. **Civilizational Collapse Risks**
    - Ecological system collapse (biodiversity loss, climate tipping points)
    - Social system collapse (institutional failure, trust breakdown)
    - Economic system collapse (financial contagion, resource scarcity)
2. **Agency Erosion Risks**
    - Cognitive hacking (AI persuasion, attention capture)
    - Democratic decay (surveillance, manipulation, elite capture)
    - Collective intelligence decline (epistemic commons degradation)
3. **Irreversibility Risks**
    - Genetic modification without recall mechanism
    - AI systems with superhuman capabilities

- Geoengineering with unknown feedback loops
- Mass extinction events

Evaluation Metrics:

- Reversibility Score (Can this be undone if it fails?)
- Resilience Impact (Does this make the system more brittle or robust?)
- Option Value (Does this preserve or foreclose future choices?)
- Threshold Proximity (How close does this bring us to critical tipping points?)

Graph Relationships:

```
Decision Node → APPROACHES / RETREATS_FROM → Tipping Point Node
            → INCREASES / DECREASES → System Resilience
            → FORECLOSES / PRESERVES → Future Option
            → CREATES → Irreversible Change
            → TRIGGERS → Cascade Effect
```

3.2 Temporal Conflict Resolution

Often, a decision will show benefits in one timescale and harms in another. The DCFM uses a **Temporal Conflict Matrix** to resolve these tensions:

```
Priority Hierarchy (in cases of conflict):

1. Deep Future Existential Risk
   - If decision creates existential risk, reject regardless of short-term benefits
   - Rationale: No amount of present benefit justifies civilizational suicide

2. Planetary Boundary Respect
   - If decision breaches ecological ceiling, reject unless it repairs greater breach
   - Rationale: Biophysical limits are non-negotiable constraints

3. Social Foundation Universality
   - If decision lifts people above social foundation, strongly favor
   - Exception: Not at expense of future generations or ecological boundaries

4. Intergenerational Debt Repair
   - If decision repairs historical harm, favor over status quo preservation
   - Exception: Not if it creates new harms to innocent parties

5. Present Optimization
   - Only after above criteria are satisfied, optimize for present benefit
   - Rationale: Present gains are meaningless if they undermine future viability
```

Temporal Discount Rate: Unlike conventional economics that discounts future impacts, the DCFM uses a **reverse temporal weighting**:

- Impacts in 0-5 years: Weight = 1.0
- Impacts in 5-25 years: Weight = 1.5
- Impacts in 25-100 years: Weight = 2.0
- Impacts beyond 100 years: Weight = 3.0

This reflects the reality that harms to future generations are ethically weightier due to:

1. Their inability to consent or participate in present decisions
2. The accumulation and amplification of impacts over time
3. The irreversibility of many long-term consequences

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4. Interstitial Impact Mapping

4.1 The Interstitial Layer: Where Complexity Lives

The **interstitials** are the most critical and most overlooked component of impact assessment. They represent:

- The gaps between plan and reality
- The emergent properties that arise from system interactions
- The unintended consequences that often dwarf intended outcomes
- The feedback loops that create path dependencies

Key Insight: In complex adaptive systems, interstitial impacts typically exceed direct impacts in magnitude and duration. The DCFM therefore dedicates equal or greater analytical resources to interstitial mapping as to direct impact assessment.

4.2 Categories of Interstitial Impacts

4.2.1 Feedback Loop Dynamics

Reinforcing Loops (Positive Feedback):

- Virtuous Cycles: Decision creates conditions that amplify beneficial outcomes
- Vicious Cycles: Decision creates conditions that amplify harmful outcomes

Balancing Loops (Negative Feedback):

- Self-Correction: System pushes back against change, maintaining homeostasis
- Resistance: Stakeholders adapt in ways that undermine intended impact

Graph Pattern:

```
Decision → Impact A → AMPLIFIES → Impact B → AMPLIFIES → Impact A
(creates reinforcing loop)

Decision → Impact A → TRIGGERS → Resistance → INHIBITS → Impact A
(creates balancing loop)
```

Example:
Decision: Implement universal basic income (UBI)

Potential Reinforcing Loop (Virtuous):
UBI → Reduced financial stress → Improved health → Reduced healthcare costs → More fiscal space for UBI → (virtuous cycle)

Potential Reinforcing Loop (Vicious):
UBI → Landlords raise rents (capture value) → Increased cost of living → Demand for higher UBI → Landlords raise rents → (vicious cycle)

Potential Balancing Loop:
UBI → Some workers exit labor market → Labor shortage → Wages increase → UBI less necessary → Political support wanes → (self-limiting)

4.2.2 Power Redistribution Effects

Core Insight: Every decision shifts power relationships, often in unexpected ways.

Power Domains to Map:

1. **Economic Power:** Control over resources and wealth flows
2. **Political Power:** Influence over collective decisions
3. **Epistemic Power:** Control over knowledge and narratives
4. **Technological Power:** Control over infrastructure and platforms
5. **Social Power:** Influence over norms and relationships

Concentration vs. Distribution:

- Does the decision concentrate power in fewer hands?
- Does it distribute power more widely?
- Does it shift power between domains (e.g., from political to economic)?

Graph Relationships:

```
Decision Node → SHIFTS_POWER_FROM → Actor A → TO → Actor B
          → CONCENTRATES_POWER → In Domain X
          → DISTRIBUTES_POWER → Across Network Y
          → CREATES_NEW_POWER_CENTER → New Actor Type
```

Example:
Decision: City adopts algorithmic traffic management system

Power Shifts:

- FROM: Local traffic engineers → TO: Software vendor (epistemic power)
- FROM: Public oversight → TO: Proprietary algorithm (political power)
- FROM: Distributed decision-making → TO: Centralized optimization (technological power)
- CREATES: New dependency on vendor for all future traffic management

4.2.3 Knowledge and Agency Erosion

Crawford's Insight: Technologies that promise to augment human capability often actually erode it through dependency and deskilling.

Types of Erosion:

1. **Cognitive Deskilling**
   - Loss of tacit knowledge through automation
   - Reduced ability to function without technological aid
   - Atrophy of judgment and intuition
2. **Social Deskilling**
   - Reduced capacity for face-to-face interaction
   - Loss of conflict resolution skills
   - Weakened community bonds
3. **Institutional Deskilling**
   - Loss of organizational memory
   - Reduced capacity for non-algorithmic decision-making
   - Dependency on external expertise
4. **Epistemic Closure**
   - Loss of indigenous or traditional knowledge
   - Devaluation of non-quantifiable ways of knowing
   - Creation of "legibility" that erases local complexity

Graph Relationships:

```
Decision Node → AUTOMATES → Human Activity
          → ERODES → Skill/Knowledge Node
          → CREATES_DEPENDENCY → On Technology
          → REDUCES → Future Capacity
          → DEVALUES → Knowledge System
```

Example:
Decision: Implement AI medical diagnosis system

Knowledge Erosion:

- Doctors lose diagnostic skills through reliance on AI recommendations
- Patients lose understanding of their own health patterns
- Medical education shifts from developing clinical judgment to managing AI tools
- Non-Western medical knowledge systems further marginalized (only biomarker-based knowledge is "legible" to AI)

4.2.4 Externality Chains

Core Insight: What institutions call "externalities" are actually interstitial impacts pushed onto commons, weaker actors, or future generations.

Externality Mapping:

1. **Spatial Externalities:** Impacts pushed to other locations (often Global South)
2. **Temporal Externalities:** Impacts pushed to future time periods
3. **Social Externalities:** Impacts pushed onto marginalized communities
4. **Ecological Externalities:** Impacts pushed onto non-human systems

The Externality Chain:

```
Decision → Private Benefit → EXTRACTS_FROM → Commons
                    → HARMS → Distant Community
                    → DEPLETES → Future Resource
                    → DEGRADES → Ecosystem
```

Example:
Decision: Manufacture smartphones using rare earth minerals

Externality Chain:

- Cobalt mining in Congo → Child labor, health hazards (social externality)
- Toxic waste from processing → Water pollution in extraction sites (ecological externality)
- E-waste exported to Ghana → Environmental and health damage (spatial externality)
- Resource depletion → Scarcity for future generations (temporal externality)
- Local knowledge disruption → Mining displaces traditional livelihoods (epistemic externality)

4.2.5 Systemic Threshold Effects

Schmachtenberger's Warning: Linear thinking misses threshold effects where systems flip to qualitatively different states.

Threshold Types:

1. **Tipping Points:** Gradual change suddenly triggers rapid, self-reinforcing shift
2. **Phase Transitions:** System moves to fundamentally different operational regime
3. **Cascades:** Failure in one domain triggers failures in connected domains
4. **Irreversibilities:** Crossing threshold makes return to previous state impossible

Graph Patterns:

```
Accumulated Impacts → APPROACHES → Threshold Node
          → CROSSES → Tipping Point
          → TRIGGERS → Cascade Event
          → CREATES → New System State
          → FORECLOSES → Return Path
```

Example:
Decision: Continue incremental increase in atmospheric CO2

Threshold Effects:

- Ocean acidification → pH drops below critical level → Mass coral die-off (tipping point)
- Arctic ice loss → Albedo shift → Accelerated warming (self-reinforcing cascade)
- AMOC circulation slowdown → Crosses threshold → Regional climate regime shift (phase transition)
- Permafrost thaw → Methane release → Runaway warming (irreversible cascade)

4.3 Interstitial Impact Assessment Protocol

For every proposed decision, the DCFM requires mapping of:

Step 1: Identify Stakeholder Network

- Who/what is directly affected?
- Who/what is connected to directly affected parties?
- Map relationships out to 3 degrees of separation

Step 2: Trace Resource Flows

- What resources are extracted, transformed, or deposited?
- Follow flows upstream (sourcing) and downstream (waste)
- Identify where flows cross boundaries (spatial, temporal, social)

Step 3: Model Feedback Dynamics

- For each direct impact, ask: "What happens next?"
- Identify reinforcing and balancing loops
- Estimate time delays in feedback

Step 4: Map Power Shifts

- Who gains decision-making capacity?
- Who loses autonomy or agency?
- What new dependencies are created?

Step 5: Assess Knowledge Impacts

- What forms of knowledge become more/less valued?
- What skills are enhanced/atrophied?
- What ways of knowing are marginalized/elevated?

Step 6: Identify Externality Chains

- What costs are not borne by the decision-maker?
- Who/what absorbs these costs?

- Are there spatial, temporal, or social displacements?

Step 7: Check for Threshold Proximity

- Are any known tipping points being approached?
- Are there non-linear responses in the system?
- What early warning signals exist?

Step 8: Evaluate Moloch Patterns

- Is this decision driven by competitive pressure?
- Does it create a race to the bottom?
- Are we trapped in escalating commitment?

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5. Governance Implementation Structure

5.1 The Wisdom Council

Purpose: A deliberative body responsible for evaluating all major institutional decisions using the DCFM framework.

Composition:

- **Elder Knowledge Holders** (25%): Indigenous elders, ecological scientists, historians
- **Affected Stakeholders** (25%): Representatives from communities directly impacted
- **Technical Experts** (20%): Systems scientists, economists, technologists
- **Future Advocates** (15%): Youth representatives, children's rights advocates
- **Non-Human Representatives** (15%): Ecological advocates, animal welfare specialists
    - Note: Explicitly includes voice for entities that cannot speak for themselves

Operating Principles:

1. **Consensus-Seeking:** Decisions require 75% agreement to proceed
2. **Precautionary Principle:** Burden of proof on those proposing change
3. **Reversibility Preference:** Favor decisions that can be undone if harmful

4. **Transparency:** All deliberations and models publicly accessible
5. **Iterative Learning:** Regular retrospectives on past decisions' actual impacts

5.2 Decision Classification System

Not all decisions require the same depth of analysis. The DCFM uses a tiered system:

Tier 1: Light Touch Review (1-2 weeks)

- Criteria: Low resource use, fully reversible, local scope, no threshold risks
- Examples: Office supply procurement, minor process changes
- Review Process: Automated DCFM screening, staff approval

Tier 2: Standard Review (1-3 months)

- Criteria: Moderate resource use, medium reversibility, regional scope
- Examples: New product launch, facility expansion, supply chain changes
- Review Process: Full DCFM analysis, Wisdom Council subcommittee

Tier 3: Deep Review (3-12 months)

- Criteria: High resource use, low reversibility, broad scope, threshold risks
- Examples: Major infrastructure, new technologies, policy overhauls
- Review Process: Comprehensive DCFM with participatory modeling, full Wisdom Council

Tier 4: Moratorium Category (Prohibited pending further governance development)

- Criteria: Existential risks, irreversible impacts, crosses known tipping points
- Examples: Self-improving AI, geoengineering, gain-of-function research
- Review Process: Cannot proceed without extraordinary consensus and global coordination

Classification Algorithm:

```
IF (Existential Risk = TRUE) OR (Irreversible = TRUE AND Threshold Crossing = TRUE)
  THEN Tier 4 (Moratorium)
ELSE IF (Resource Impact = HIGH) OR (Reversibility = LOW) OR (Scope = GLOBAL)
  THEN Tier 3 (Deep Review)
ELSE IF (Resource Impact = MEDIUM) OR (Scope = REGIONAL)
  THEN Tier 2 (Standard Review)
ELSE
  Tier 1 (Light Touch)
```

5.3 Charter Framework Components

Every institution operating under DCFM governance must adopt a charter containing:

1. Do No Harm Principle "This institution acknowledges its fiduciary duty to life—human and non-human, present and future. We commit to evaluating all decisions through the lens of minimizing harm across all timescales and all affected parties, including those without voice in current power structures."

2. Doughnut Commitment "We commit to operating within the 'safe and just space' defined by social foundations and planetary boundaries. We will not pursue institutional goals that require breaching ecological ceilings or leaving human needs unmet."

3. Transparency Mandate "We commit to making our DCFM analysis publicly available, including our modeling assumptions, identified uncertainties, and dissenting views within our Wisdom Council."

4. Precautionary Application "In cases of uncertainty about potential harms, we place burden of proof on the proposed action. We favor reversible interventions and iterative learning over large-scale, irreversible changes."

5. Stakeholder Inclusion "We commit to including in our governance process: affected communities, future generation advocates, non-human representatives, and holders of marginalized knowledge systems."

6. Moloch Awareness "We acknowledge the risk of competitive dynamics driving harmful decisions. We commit to seeking cooperative solutions and supporting regulatory frameworks that prevent races to the bottom."

7. Reparative Justice "We acknowledge historical harms perpetuated by extractive models. We commit to directing surplus toward repairing damage to communities, ecosystems, and knowledge systems harmed by past exploitation."

8. Living Document Status "This charter is a living document, subject to revision as we learn from our decisions' actual impacts and as the state of collective knowledge evolves."

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6. Graph-Based Representation

6.1 Technical Architecture

The DCFM is implemented as a **property graph database** with the following structure:

Node Types:

- `Decision`: Proposed institutional action
- `Impact`: Consequence of decision (direct, interstitial, or recursive)
- `Actor`: Human, institutional, or non-human entity
- `Resource`: Natural, human, or manufactured capital
- `Boundary`: Social foundation or planetary boundary threshold
- `Knowledge`: Information, skill, or wisdom system
- `Feedback`: Loop or cascade pattern
- `Risk`: Identified hazard or vulnerability

Edge Types (with properties):

- `CAUSES [strength, confidence, time_delay]`
- `ENABLES [mechanism, context]`
- `HARMS [magnitude, reversibility, affected_parties]`
- `BENEFITS [magnitude, distribution, duration]`
- `SHIFTS_POWER [from, to, domain]`
- `ERODES [what, rate, threshold]`
- `APPROACHES [threshold_distance, trajectory]`
- `CREATES_DEPENDENCY [on, reversibility]`

Property Examples:

```
json
```

```
Decision Node {
  "id": "dec_001",
  "title": "Deploy AI traffic system",
  "proposing_actor": "city_transport_dept",
  "date_proposed": "2025-01-15",
  "tier": 2,
  "status": "under_review"
}

Impact Node {
  "id": "imp_023",
  "type": "interstitial",
  "description": "Erosion of human traffic engineer skills",
  "category": "knowledge_erosion",
  "magnitude": "moderate",
  "reversibility": "low",
  "time_horizon": "5-10 years"
}

Edge {
  "from": "dec_001",
  "to": "imp_023",
  "type": "CAUSES",
  "strength": 0.8,
  "confidence": 0.6,
  "time_delay": "2 years",
  "mechanism": "automation_deskilling"
}
```

6.2 Graph Algorithms for Analysis

1. Impact Propagation Algorithm Traces effects through the graph to identify cascades and feedback loops.

```python
```

```python
def propagate_impact(decision_node, max_depth=5, time_horizon=10):
    """
    Traces impact chains from a decision through the causal network.
    Returns: List of (impact_node, path, cumulative_magnitude) tuples
    """
    impacts = []
    visited = set()

    def dfs(current_node, path, cumulative_magnitude, depth, time_elapsed):
        if depth > max_depth or time_elapsed > time_horizon:
            return

        visited.add(current_node)

        for edge in current_node.outgoing_edges:
            if edge.type in ['CAUSES', 'TRIGGERS', 'ENABLES', 'AMPLIFIES']:
                next_node = edge.target
                new_magnitude = cumulative_magnitude * edge.strength
                new_time = time_elapsed + edge.time_delay
                new_path = path + [edge]

                if next_node.type == 'Impact':
                    impacts.append((next_node, new_path, new_magnitude))

                if next_node not in visited:
                    dfs(next_node, new_path, new_magnitude, depth+1, new_time)

    dfs(decision_node, [], 1.0, 0, 0)
    return impacts
```

2. Moloch Pattern Detector

```python
```

```python
def detect_moloch_patterns(decision_node):
    """
    Identifies competitive dynamics driving harmful collective outcomes.
    Returns: List of detected Moloch patterns
    """
    patterns = []

    # Pattern 1: Race to the Bottom
    competitors = find_competitors(decision_node.proposing_actor)
    if has_competitive_pressure(decision_node, competitors):
        collective_outcome = assess_collective_impact(decision_node, competitors)
        if collective_outcome.net_harm > threshold:
            patterns.append({
                'type': 'race_to_bottom',
                'competitors': competitors,
                'individual_incentive': 'positive',
                'collective_outcome': 'negative'
            })

    # Pattern 2: Externality Export
    beneficiaries = find_beneficiaries(decision_node)
    harm_bearers = find_harm_bearers(decision_node)
    if not set(beneficiaries).intersection(set(harm_bearers)):
        patterns.append({
            'type': 'externality_export',
            'beneficiaries': beneficiaries,
            'harm_bearers': harm_bearers
        })

    return patterns
```

3. Doughnut Compliance Checker

```
python
```

```python
def check_doughnut_compliance(decision_node):
    """
    Evaluates whether decision respects social foundations and ecological ceilings.
    Returns: Compliance score and violated boundaries
    """
    social_foundation_impacts = {}
    ecological_ceiling_impacts = {}

    for dimension in SOCIAL_FOUNDATION_DIMENSIONS:
        impact = assess_impact_on_dimension(decision_node, dimension)
        social_foundation_impacts[dimension] = impact

    for boundary in PLANETARY_BOUNDARIES:
        pressure = assess_pressure_on_boundary(decision_node, boundary)
        ecological_ceiling_impacts[boundary] = pressure

    violations = []

    # Check social foundation shortfalls
    for dim, impact in social_foundation_impacts.items():
        if impact < 0:  # Worsens access to social need
            violations.append(f"Degrades {dim} (social foundation)")

    # Check ecological ceiling breaches
    for boundary, pressure in ecological_ceiling_impacts.items():
        current_state = get_current_boundary_state(boundary)
        if current_state + pressure > boundary.threshold:
            violations.append(f"Breaches {boundary.name} (planetary boundary)")

    compliance_score = calculate_doughnut_score(
        social_foundation_impacts,
        ecological_ceiling_impacts
    )

    return compliance_score, violations
```

4. Temporal Risk Analyzer

```python
```

```python
def analyze_temporal_risks(decision_node):
    """
    Evaluates decision across three timescales with reverse weighting.
    Returns: Weighted risk score across timescales
    """
    risks = {
        'intergenerational_past': assess_historical_debt(decision_node),
        'present_doughnut': assess_present_impacts(decision_node),
        'deep_future': assess_existential_risks(decision_node)
    }

    # Apply reverse temporal weighting
    weights = {
        'intergenerational_past': 1.5,
        'present_doughnut': 1.0,
        'deep_future': 3.0
    }

    weighted_score = sum(
        risks[timescale] * weights[timescale]
        for timescale in risks
    )

    # Check for existential red flags
    if risks['deep_future'].existential_risk:
        return {'status': 'PROHIBITED', 'reason': 'Existential risk detected'}

    return {
        'status': 'REQUIRES_REVIEW' if weighted_score > threshold else 'APPROVED',
        'weighted_score': weighted_score,
        'timescale_breakdown': risks
    }
```

6.3 Visualization and Communication

The graph structure enables multiple visualization modes for different audiences:

1. Decision Impact Network Shows decision at center with radiating impacts, color-coded by type and magnitude.

2. Stakeholder Perspective View Shows impacts from the perspective of a specific stakeholder group (e.g., "future generations" view, "ecosystem" view).

3. Temporal Flow Diagram Displays how impacts unfold over time, with branches showing different scenarios.

4. Threshold Dashboard Shows proximity to critical planetary and social boundaries, with decision's contribution highlighted.

5. Power Flow Diagram Visualizes shifts in power and control resulting from decision.

---

7. Decision Evaluation Protocol

7.1 Standard Operating Procedure

Phase 1: Pre-Submission (Institution's Responsibility)

1. **Initial DCFM Scan**
   - Run automated impact propagation analysis
   - Generate preliminary stakeholder map
   - Identify obvious Moloch patterns or threshold risks
2. **Tier Classification**
   - Determine appropriate review tier
   - Estimate timeline and resources needed
3. **Stakeholder Notification**
   - Identify and notify all parties within 2 degrees of separation
   - Provide 30-day comment period for Tier 2+
   - Provide 90-day comment period for Tier 3

Phase 2: Wisdom Council Review

4. **Model Construction**
   - Build full causal graph for decision
   - Incorporate stakeholder input and concerns
   - Identify uncertainties and knowledge gaps
5. **Multi-Perspective Analysis**
   - Run all algorithmic checks (Moloch, Doughnut, Temporal)
   - Conduct qualitative assessment from multiple viewpoints

- Engage specialized expertise as needed

6. **Deliberation**
   - Present findings to full Wisdom Council
   - Debate tradeoffs and alternatives
   - Consider precautionary principle application
7. **Decision**
   - Approve, Reject, or Request Modifications
   - Document reasoning and dissenting views
   - Publish full DCFM analysis

Phase 3: Post-Implementation

8. **Monitoring**
   - Track actual impacts vs. predicted impacts
   - Watch for emergence of unpredicted interstitials
   - Maintain alert for threshold approaches
9. **Retrospective**
   - Annual review of decision's effects
   - Update models based on learned realities
   - Adjust decision if harms materialize
10. **Knowledge Integration**
    - Feed learnings back into DCFM database
    - Refine algorithms and heuristics
    - Share insights with broader governance network

7.2 Evaluation Criteria and Scoring

Each decision receives scores across multiple dimensions:

A. Doughnut Alignment Score (-12 to +12)

- Social Foundation: -12 to 0 (harms) or 0 to +12 (benefits) across 12 dimensions
- Ecological Ceiling: -9 to 0 (breaches) or 0 to +9 (respects) across 9 boundaries

B. Temporal Responsibility Score (0 to 100)

- Historical Debt Repair: 0-25 points
- Present Generation Wellbeing: 0-25 points
- Future Generation Safeguarding: 0-50 points (double weighted)

C. Power Distribution Score (-10 to +10)

- Negative: Concentrates power, creates dependencies
- Positive: Distributes power, enhances autonomy

D. Knowledge Impact Score (-10 to +10)

- Negative: Erodes skills, marginalizes knowledge systems
- Positive: Enhances capacity, honors diverse epistemologies

E. Resilience Score (-10 to +10)

- Negative: Increases brittleness, creates single points of failure
- Positive: Enhances adaptive capacity, maintains redundancy

F. Moloch Vulnerability Score (0 to 100)

- 0-25: Not driven by competitive pressure
- 26-50: Some competitive elements, manageable
- 51-75: Significant Moloch patterns, concerning
- 76-100: Fully trapped in race to bottom, reject

Aggregate Approval Logic:

```
IF Moloch Score > 75 OR Existential Risk = TRUE OR Doughnut Score < -5
  THEN REJECT
ELSE IF Temporal Score < 40 OR Resilience Score < -5
  THEN REQUIRE MODIFICATIONS
ELSE IF All Scores within acceptable range AND Stakeholder Consensus >= 75%
  THEN APPROVE
ELSE
  THEN RETURN TO DELIBERATION
```

---

8. Charter Framework: Do No Harm

8.1 Institutional Charter Template

```
[Institution Name] Do No Harm Charter
Adopted: [Date]
Version: [X.X]

PREAMBLE

We, [institution name], recognize that we operate within nested systems—ecological, social, economic, and epistemic—that sustain all life and possibility. We acknowledge that past models of institutional success have often been predicated on externalizing harms onto communities, ecosystems, and future generations who lack power to resist.

We commit to a fundamental reorientation: from extraction to regeneration, from exploitation to emancipation, from rivalry to collaboration, from efficiency to wisdom.

This charter represents our binding commitment to evaluate every decision through the Dynamic Causal Factor Model (DCFM), ensuring that we do no harm to people or planet across all timescales.

ARTICLE I: FOUNDATIONAL COMMITMENTS

Section 1.1: Fiduciary Duty to Life
We acknowledge our primary fiduciary duty is to life itself—human and non-human, present and future. This duty supersedes all other institutional objectives, including profit maximization, market share growth, and competitive advantage.

Section 1.2: Boundary Respect
We commit to operating within planetary boundaries and above social foundations, as defined by the Doughnut Economics framework. We will not pursue goals that require breaching ecological ceilings or leaving human needs systematically unmet.
```

Section 1.3: Temporal Justice

We acknowledge obligations to:

- Past generations whose knowledge and sacrifices enabled our present capacity
- Present generations whose wellbeing depends on our wise stewardship
- Future generations who will inherit the consequences of our decisions

We apply reverse temporal weighting, giving greater consideration to impacts that unfold over longer time horizons.

Section 1.4: Interconnection

We acknowledge that no decision is isolated. Every action ripples through networks of relationship. We commit to mapping and minimizing negative externalities while enhancing positive spillovers.

ARTICLE II: GOVERNANCE STRUCTURES

Section 2.1: Wisdom Council

We establish a Wisdom Council with the composition and authority outlined in Section 5 of the DCFM framework. This Council has veto power over decisions that violate Do No Harm principles.

Section 2.2: Stakeholder Rights

We recognize rights of participation for:

- Communities directly affected by our operations
- Representatives of future generations
- Advocates for non-human entities
- Holders of marginalized knowledge systems

Section 2.3: Transparency

We commit to making public:

- All DCFM analyses for Tier 2+ decisions
- Our Wisdom Council deliberations and votes
- Our monitoring data on actual vs. predicted impacts
- Our retrospective learnings and model updates

ARTICLE III: DECISION EVALUATION

Section 3.1: Mandatory DCFM Application

All institutional decisions above minimal threshold must undergo DCFM evaluation, including:

- Impact propagation analysis across three timescales
- Interstitial mapping including power shifts and knowledge impacts
- Moloch pattern detection
- Doughnut compliance checking
- Existential risk assessment

Section 3.2: Precautionary Principle

In cases of scientific uncertainty about potential harms, the burden of proof rests on the proposed action. We favor:
- Reversible interventions over irreversible
- Small-scale experiments over large-scale rollouts
- Adaptive management over rigid planning

Section 3.3: Prohibition Categories
We will not pursue decisions that:
- Create existential risks to human civilization
- Cross known planetary tipping points
- Permanently foreclose future generations' options
- Systematically erode human agency or cognitive capacity
- Depend on competitive races to the bottom

ARTICLE IV: OPERATIONAL PRINCIPLES

Section 4.1: Regenerative Design
We commit to designing our operations to restore rather than degrade:
- Natural systems: Enhance ecosystem health and biodiversity
- Social systems: Build trust, cooperation, and collective capacity
- Knowledge systems: Honor and integrate diverse epistemologies

Section 4.2: Distributive Justice
We commit to ensuring that:
- Benefits of our activities are widely distributed
- Burdens are not disproportionately borne by vulnerable populations
- Wealth flows support rather than extract from communities

Section 4.3: Collaborative Over Competitive
We seek cooperative solutions and support regulatory frameworks that prevent races to the bottom. We will not use competitive pressure as justification for harmful practices.

Section 4.4: Knowledge Sovereignty
We respect the rights of communities to control their own knowledge, particularly:
- Indigenous knowledge of ecological relationships
- Traditional agricultural and medicinal practices
- Local innovations and adaptations

We will not engage in biopiracy or the appropriation of traditional knowledge without free, prior, and informed consent and equitable benefit-sharing.

ARTICLE V: ACCOUNTABILITY AND ADAPTATION

Section 5.1: Monitoring and Reporting
We commit to:

- Real-time monitoring of our ecological and social impacts
- Annual public reporting on our Doughnut compliance
- Transparent disclosure of our proximity to critical thresholds

Section 5.2: Adaptive Response
When our monitoring reveals unanticipated harms, we will:
- Immediately halt activities crossing red lines
- Engage affected stakeholders in designing responses
- Implement corrective measures within [specified timeframe]
- Compensate harmed parties to the extent possible

Section 5.3: Continuous Improvement
We treat this charter as a living document, subject to revision as:
- Scientific understanding of planetary boundaries evolves
- Communities identify new categories of harm
- We learn from our decision outcomes
- Best practices in regenerative governance emerge

Section 5.4: Sunset Clause
If we consistently fail to honor these commitments, or if our operational model proves incompatible with Do No Harm principles, we commit to orderly dissolution and transfer of resources to institutions better aligned with life-affirming values.

ARTICLE VI: ENFORCEMENT

Section 6.1: Internal Enforcement
- Wisdom Council has authority to override management decisions
- Any member of the institution can trigger DCFM review
- Whistleblower protections for those identifying charter violations

Section 6.2: External Accountability
We invite oversight from:
- Affected community representatives
- Independent DCFM auditors
- Regulatory bodies aligned with regenerative principles

Section 6.3: Remedy Mechanisms
We establish clear pathways for:
- Grievance filing by affected parties
- Binding arbitration of disputes
- Restorative justice processes

ADOPTION

This charter is adopted by [governance body] on [date] and takes effect immediately. It supersedes any prior institutional commitments that conflict with these principles.

Signed:
[Institutional Leaders]
[Wisdom Council Representatives]
[Community Stakeholders]

8.2 Charter Implementation Checklist

For institutions adopting the Do No Harm Charter:

Immediate Actions (Month 1)

- [ ] Establish interim Wisdom Council
- [ ] Conduct institutional audit of current decision-making processes
- [ ] Identify decisions currently in pipeline requiring DCFM review
- [ ] Begin stakeholder mapping for major operations
- [ ] Publish charter and solicit public comment

Short-Term Actions (Months 2-6)

- [ ] Build DCFM graph database infrastructure
- [ ] Train staff in DCFM methodology
- [ ] Develop tier classification protocols
- [ ] Establish monitoring systems for key impacts
- [ ] Conduct first retrospective analysis of past decisions

Medium-Term Actions (Months 7-18)

- [ ] Complete DCFM analysis of all Tier 3 legacy operations
- [ ] Implement modifications to achieve Doughnut compliance
- [ ] Establish public dashboard for transparency
- [ ] Build relationships with affected communities
- [ ] Create knowledge-sharing networks with other charter institutions

Long-Term Actions (18+ months)

- [ ] Achieve full operational alignment with charter principles
- [ ] Document learnings and refine DCFM models
- [ ] Support ecosystem of regenerative institutions
- [ ] Advocate for regulatory frameworks preventing Moloch traps

9. Practical Implementation Roadmap

9.1 Roadmap for Local Community (City/Region)

Case Study: Mid-Sized City (Portland, Oregon or Bristol, UK Model)

Phase 1: Foundation (Months 1-6) - "The Local Audit"

Goal: Establish baseline understanding and governance infrastructure

Actions:

1. **Form the Transition Alliance**
    - Convene diverse stakeholders: city government, community organizations, labor unions, indigenous representatives, youth councils, ecological scientists
    - Establish interim Wisdom Council
    - Secure initial funding for transition process
2. **Conduct Multi-Capital Inventory**
    - Map natural capital: local ecosystems, water systems, agricultural land, biodiversity
    - Map human capital: local knowledge systems, skills, social networks, cultural practices
    - Map manufactured capital: infrastructure, buildings, technology systems
    - Map financial flows: where does wealth come from and where does it go?
3. **Doughnut Audit**
    - Assess social foundation: Identify which populations lack access to:
        - Food security (map food deserts)
        - Housing (assess affordability crisis)
        - Healthcare (identify gaps in coverage)
        - Education (map quality disparities)
        - Income and work (unemployment, precarity, wage adequacy)
        - Energy (energy poverty, access to clean energy)
    - Assess ecological ceiling: Calculate city's footprint on:
        - Climate (GHG emissions, per capita carbon budget)
        - Biodiversity (urban habitat loss, species decline)

- Water (consumption, pollution, watershed health)
- Materials (waste generation, recycling rates)
- Air quality (particulate matter, pollutants)

4. **Imperial Dependency Analysis**
   - Map global supply chains (where does food, energy, goods come from?)
   - Identify extractive relationships (are we benefiting from Global South exploitation?)
   - Assess technofeudal dependencies (which platforms and corporations control local digital infrastructure?)
   - Calculate wealth transfer (how much capital flows out vs. stays local?)
5. **Risk and Resilience Assessment**
   - Identify vulnerabilities to climate shocks
   - Map single points of failure in critical systems
   - Assess social cohesion and trust levels
   - Evaluate adaptive capacity

Deliverable: Comprehensive baseline report with DCFM graph database populated with current state, publicly accessible dashboard

Phase 2: Building Earth Democracy (Months 7-18) - "The Solidarity Infrastructure"

Goal: Create alternative institutions that embody regenerative principles

Actions:

1. **Establish Seed Sovereignty**
   - Create community seed bank with focus on locally-adapted varieties
   - Map and protect heirloom seed knowledge held by elder gardeners
   - Establish seed-saving workshops and knowledge transfer
   - Pass municipal ordinance protecting seed saving rights (vs. corporate patent claims)
   - Support urban agriculture using saved seeds
2. **Build Cooperative Economy**
   - Launch cooperative conversion fund to help businesses transition to worker ownership
   - Establish platform cooperatives for local services (ride-sharing, delivery, care work)
   - Create cooperative incubator with legal/financial support

- Implement "Doughnut Procurement" policy:
    - Give preference to cooperatives in city contracts
    - Require bidders to disclose Doughnut scores
    - Award contracts based on social + ecological value, not just lowest price

3. **Create Public Banking Option**
    - Establish municipal bank modeled on public banks in Germany and North Dakota
    - Capitalize with public funds and community investment
    - Mission: finance Doughnut-aligned projects
    - Offer:
        - Low-interest loans for cooperatives
        - Green mortgages tied to energy efficiency
        - Small business loans with social mission criteria
    - Keep wealth circulating locally vs. extracting to Wall Street
4. **Implement Participatory Budgeting**
    - Allocate portion of city budget (start with 5-10%) to participatory process
    - Use DCFM framework to evaluate citizen proposals
    - Create "Sense-Making Assemblies" where residents deliberate on tradeoffs
    - Weight input from youth and affected communities
5. **Establish Digital Commons**
    - Build municipal mesh network for internet access
    - Create open-source alternatives to proprietary platforms
    - Develop local digital currency/mutual credit system
    - Ensure data sovereignty (data stays local, under community control)
    - Reduce dependencies on Big Tech surveillance infrastructure

Deliverable: Functional solidarity economy infrastructure, measurable reduction in imperial dependencies, increased local wealth circulation

Phase 3: Systemic Transformation (Months 19-36) - "The Regenerative Pivot"

Goal: Achieve fundamental reorientation from extractive to regenerative model

Actions:

1. **Bioregional Food System**
    - Establish "Solidarity Pacts" with surrounding rural areas
    - Create guaranteed markets for regenerative farmers
    - Build community-owned food processing infrastructure
    - Close the loop: compost food waste back to farms
    - Target: 50% of food consumed locally produced by regenerative methods
    - Metrics: Soil health improvement, reduced pesticide use, farmer livelihoods
2. **Regenerative Energy**
    - Community-owned solar and wind projects
    - Retrofit buildings for energy efficiency (prioritizing low-income housing)
    - Implement "Green New Deal" job guarantee for energy transition work
    - Phase out fossil fuel infrastructure
    - Target: 100% renewable energy, locally owned
3. **Circular Materials Economy**
    - Ban single-use plastics
    - Establish repair cafes and tool libraries
    - Create materials exchange for construction waste
    - Support businesses designing for durability and repairability
    - Target: 90% waste diversion from landfill
4. **Ecological Restoration**
    - Rewild portions of urban area (daylight streams, plant native species)
    - Remove impervious surfaces, restore water infiltration
    - Create wildlife corridors
    - Ban pesticides on public land
    - Target: Net gain in urban biodiversity, improved ecosystem services
5. **Care Economy Investment**
    - Expand public childcare and elder care
    - Pay care workers living wages
    - Support neighborhood mutual aid networks
    - Recognize and value unpaid care work

- Target: No care deserts, reduced inequality in care burden

6. **Retire GDP as Success Metric**
   - Officially adopt Genuine Progress Indicator (GPI) or Doughnut Dashboard
   - Report annually on progress toward thriving within boundaries
   - Celebrate wellbeing improvements, not just economic growth
   - Redefine "prosperity" in terms of life quality, not consumption

Deliverable: Transformation visible in lived experience, measurable improvement on Doughnut indicators, model replicable by other cities

Phase 4: Scaling the Model (Year 3+) - "The Network Effect"

Goal: Build resilient network of regenerative localities, advocate for policy change

Actions:

1. **Inter-City Solidarity Network**
   - Form alliances with other cities implementing similar models
   - Share learnings, tools, and best practices
   - Create mutual aid agreements for climate resilience
   - Build collective bargaining power vs. corporate and national government
2. **Regional Coordination**
   - Work with adjacent municipalities to align policies
   - Build bioregional governance structures
   - Coordinate on watershed management, habitat corridors, transportation
   - Create regional circular economy (what one locality discards, another uses)
3. **Policy Advocacy**
   - Push for national/international frameworks that support regenerative localities
   - Advocate for regulations preventing Moloch traps (e.g., mandatory supply chain transparency, prohibition on planned obsolescence)
   - Support carbon fee and dividend policies
   - Demand Global South debt cancellation and reparations
4. **Knowledge Sharing**
   - Document transition process in detail

- Create open-source toolkit for other communities
- Host learning exchanges and site visits
- Contribute to global movement for economic democracy

5. **Long-Term Resilience**

- Continue adapting to climate impacts
- Maintain social cohesion through challenging transitions
- Protect gains against backlash or co-optation
- Build inter-generational transmission of regenerative values

Key Performance Indicators (KPIs) for the Roadmap:

Domain	Baseline	Year 1 Target	Year 3 Target	Year 5 Target
Social Foundation				
Housing affordability	45% cost-burdened	40%	30%	<20%
Food security	15% food insecure	12%	8%	<5%
Income equality (Gini)	0.45	0.43	0.38	<0.35
Ecological Ceiling				
Per capita emissions (tons CO2)	12	10	6	<3
Waste to landfill (% of total)	70%	55%	30%	<10%
Pesticide use (kg/hectare)	8	6	2	0
Economic Transformation				
Local ownership (% businesses)	20%	30%	50%	70%
Wealth circulation (% stays local)	45%	55%	70%	85%
Cooperative employment	5%	10%	20%	35%
Resilience				
Food self-sufficiency	10%	20%	40%	60%
Energy self-sufficiency	5%	15%	50%	100%
Social trust index	45/100	50	60	70

9.2 Roadmap for Corporations/Large Institutions

Case Study: Large Corporation Transitioning to Do No Harm Charter

Year 1: Assessment and Foundation

Q1: Internal Audit

- Conduct full DCFM analysis of current operations
- Map all supply chains using participatory methods (include suppliers' workers)
- Calculate true cost accounting (internalize all externalities)

- Identify major harm areas and quick wins
- Establish interim Wisdom Council with external stakeholders

Q2: Stakeholder Engagement

- Identify all affected parties (workers, communities, ecosystems)
- Conduct listening sessions with marginalized voices
- Map power relationships and dependencies
- Begin reparative justice conversations with historically harmed communities
- Publish transparency reports

Q3: Charter Adoption and Governance Shift

- Draft and adopt Do No Harm Charter
- Restructure board to include stakeholder representatives
- Implement DCFM decision protocols
- Train leadership and staff in new framework
- Establish monitoring and accountability systems

Q4: Operational Redesign

- Phase out most harmful practices (based on Q1 audit)
- Launch pilot regenerative projects
- Begin supply chain transformation (prioritize cooperatives and regenerative suppliers)
- Implement living wage across all operations
- Start measuring Doughnut indicators

Year 2-3: Transformation

- Achieve Doughnut compliance across all operations
- Transition to circular business models
- Build long-term relationships with regenerative suppliers
- Invest surplus in community wealth-building
- Support policy changes that enable others to follow

Year 3+: Leadership

- Become vocal advocate for systemic change
- Share learnings and tools openly
- Support cooperative conversion of industry peers
- Engage in political advocacy for regulations that prevent races to bottom
- Consider cooperative conversion or steward-ownership model for own company

Major Challenges and Mitigation:

Challenge	Mitigation Strategy
Shareholder pressure for profits	Educate on long-term value; transition to stakeholder ownership model
Competitive disadvantage	Advocate for regulatory floor; build consumer demand for regenerative products
Supply chain complexity	Start with high-impact areas; use transparency to build trust
Loss of "efficiency"	Redefine efficiency to include externalities; measure true cost
Cultural resistance internally	Engage employees in participatory redesign; demonstrate benefits

---

10. Appendices

Appendix A: Ontological Schemas in Machine-Readable Format

JSON-LD Schema for Decision Node:

```
json
```

```
{
  "@context": {
    "@vocab": "http://dcfm.org/schema/",
    "dc": "http://purl.org/dc/terms/",
    "xsd": "http://www.w3.org/2001/XMLSchema#"
  },
  "@type": "Decision",
  "@id": "decision:001",
  "dc:title": "Implement AI hiring system",
  "dc:description": "Deploy algorithmic screening for job candidates",
  "proposedBy": {
    "@type": "InstitutionalActor",
    "@id": "actor:hr_dept",
    "name": "Human Resources Department"
  },
  "dateProposed": "2025-01-15T00:00:00Z",
  "reviewTier": 2,
  "status": "under_review",
  "directImpacts": [
    {
      "@type": "Impact",
      "@id": "impact:001",
      "description": "Reduce time-to-hire by 40%",
      "magnitude": 0.8,
      "category": "efficiency_gain"
    }
  ],
  "interstitialImpacts": [
    {
      "@type": "Impact",
      "@id": "impact:002",
      "description": "Algorithmic bias perpetuates discrimination",
      "magnitude": -0.7,
      "category": "social_harm",
      "affectedParties": ["marginalized_candidates"],
      "reversibility": "low"
    },
    {
      "@type": "Impact",
      "@id": "impact:003",
      "description": "HR staff deskilling",
      "magnitude": -0.5,
      "category": "knowledge_erosion",
```

```
      "timeHorizon": "5-10 years",
      "reversibility": "medium"
    }
  ],
  "recursiveImpacts": [
    {
      "@type": "Impact",
      "@id": "impact:004",
      "description": "Reduced institutional capacity for human judgment in hiring",
      "magnitude": -0.6,
      "category": "capacity_reduction",
      "timeHorizon": "10+ years"
    }
  ],
  "doughnutAssessment": {
    "socialFoundationScore": -2,
    "violatedDimensions": ["social_equity"],
    "ecologicalCeilingScore": -1,
    "breachedBoundaries": ["chemical_pollution"],
    "explanation": "Energy use for AI training and operation contributes to carbon emissions"
  },
  "molochPatterns": [
    {
      "type": "efficiency_paradox",
      "description": "Optimization for hiring speed degrades human judgment capacity"
    }
  ],
  "temporalAnalysis": {
    "intergenerationalPast": {
      "score": 0,
      "notes": "Does not directly address historical harms"
    },
    "presentDoughnut": {
      "score": -3,
      "notes": "Harms social equity, breaches ecological boundary"
    },
    "deepFuture": {
      "score": -5,
      "notes": "Contributes to agency erosion and algorithmic dependence"
    },
    "weightedTotal": -16.5
  },
  "recommendation": "REQUIRES_MODIFICATION",
  "suggestedModifications": [
```

```
    "Implement human-in-the-loop with override authority",
    "Conduct regular bias audits with affected community participation",
    "Maintain human hiring capacity through parallel non-algorithmic process",
    "Use renewable energy for computation",
    "Limit data retention and ensure data sovereignty"
  ]
}
```

Appendix B: Case Study Examples

Case Study 1: Municipal Composting Program

Decision: Implement city-wide composting collection

DCFM Analysis:

Direct Impacts:

- Reduce waste to landfill by 30%
- Create 50 green jobs
- Reduce methane emissions

Interstitial Impacts:

- Improved soil health in urban agriculture projects (positive feedback loop)
- Educational opportunities in schools (knowledge enhancement)
- Reduced need for chemical fertilizers (breaks dependency)
- Community participation increases social cohesion (power distribution)

Recursive Impacts:

- Citizens develop ecological literacy
- City builds capacity for other circular economy initiatives
- Creates model for other municipalities

Temporal Analysis:

- Past: Partially repairs damage from historical waste mismanagement
- Present: Strong Doughnut alignment (reduces ecological ceiling pressure, supports food security)

- Future: No significant risks, enhances resilience

Moloch Check: No competitive pressure driving decision; purely beneficial

Assessment: APPROVED - Exemplar of regenerative policy

Case Study 2: Smart City Surveillance System

Decision: Deploy AI-powered surveillance cameras throughout city

DCFM Analysis:

Direct Impacts:

- Claimed reduction in crime rates
- Faster emergency response times

Interstitial Impacts:

- Massive power shift from citizens to government/tech vendor (concentration)
- Chilling effect on protest and free assembly (erodes political agency)
- Disproportionate surveillance of marginalized communities (social equity harm)
- Creates dependency on proprietary technology (vendor lock-in)
- Normalizes surveillance culture (recursive impact on expectations of privacy)

Recursive Impacts:

- Erodes democratic norms and trust
- Trains population to accept loss of privacy
- Reduces collective capacity for civil disobedience
- Makes future transitions away from surveillance state more difficult

Temporal Analysis:

- Past: Echoes historical patterns of state monitoring of marginalized groups
- Present: Violates social equity dimension of Doughnut (social foundation)
- Future: High risk of authoritarian slide (Harari's "digital dictatorship")

Moloch Check: Potential race to bottom as cities feel pressure to "keep up" with neighbors' surveillance capacity

Existential Risk: Contributes to Harari's "hacking humans" trajectory

Assessment: REJECTED - Fails Do No Harm test on multiple grounds

Recommended Alternative: Invest in community-led public safety initiatives, restorative justice, addressing root causes of crime (housing, employment, mental health)

Case Study 3: Worker Cooperative Conversion

Decision: Corporation transitions to worker ownership model

DCFM Analysis:

Direct Impacts:

- Workers gain equity stake
- Democratize workplace decision-making
- Potential reduction in profit extraction

Interstitial Impacts:

- Power redistribution from distant shareholders to local workers (major positive shift)
- Increased worker engagement and well-being
- Wealth kept in local community (breaks extraction pattern)
- Inspires other businesses to consider similar transition
- Potential supply chain improvements as workers choose values-aligned partners

Recursive Impacts:

- Workers develop governance capacity and business literacy
- Organization becomes more adaptive (workers have full information)
- Creates resilience against hostile takeovers or financialization

Temporal Analysis:

- Past: Repairs historical pattern of labor exploitation
- Present: Strongly aligned with Doughnut (improves income equality, maintains ecological responsibility)

- Future: Increases long-term sustainability, preserves future options

Moloch Check: Removes competitive pressure to maximize short-term returns

Assessment: STRONGLY APPROVED - Model of emancipatory transformation

Implementation Support: Provide technical assistance, legal resources, financing for worker buyout

Appendix C: Glossary of Key Terms

Agency Erosion: The gradual reduction in human capacity for autonomous decision-making and action, often through technological dependency or cognitive deskilling.

Doughnut Economics: Framework developed by Kate Raworth defining a "safe and just space" between social foundations (minimum needs) and ecological ceilings (planetary boundaries).

Earth Democracy: Vandana Shiva's concept of governance that recognizes rights of all species and protects commons from privatization.

Externality: In economics, a cost or benefit affecting parties who did not choose to incur it. In DCFM, externalities are reframed as interstitial impacts on stakeholders without decision-making power.

Interstitials: The emergent, unintended impacts that arise in the "gaps" between planned actions and system responses; often more significant than direct impacts.

Metacrisis: Daniel Schmachtenberger's term for the interconnected web of crises (ecological, social, epistemic) whose solutions often worsen other dimensions of the crisis.

Moloch: Ancient deity used metaphorically to represent competitive dynamics that force all actors into collectively harmful behavior (race to bottom).

Planetary Boundaries: Nine biophysical thresholds identified by Earth system scientists (Rockström et al.) beyond which Earth system stability is threatened.

Recursive Impact: How a decision changes the decision-maker's future capacity, creating feedback loops on institutional or human capability.

Technofeudalism: Yanis Varoufakis's term for the emerging system where digital platforms extract rent and control, resembling feudal relationships.

Temporal Mismatch: When benefits of a decision accrue to one time period (often present) while harms fall on another (often future).

Third Attractor: Alternative to the binary of authoritarianism vs. chaos; a regenerative governance model based on wisdom and stewardship.

Appendix D: Further Reading and Resources

Core Texts:

1. **Atlas of AI** by Kate Crawford - On digital colonialism and material costs of AI
2. **Doughnut Economics** by Kate Raworth - Framework for thriving within boundaries
3. **The Entrepreneurial State** by Mariana Mazzucato - On public sector's role in innovation
4. **Mission Economy** by Mariana Mazzucato - On moonshot approach to grand challenges
5. **Progressive Capitalism** by Joseph Stiglitz - On balancing markets with social good
6. **Technofeudalism** by Yanis Varoufakis - On platform capitalism and digital rents
7. **Earth Democracy** by Vandana Shiva - On resisting biopiracy and protecting commons
8. **Sapiens** and **Homo Deus** by Yuval Noah Harari - On human history and future risks
9. **The Patterning Instinct** by Jeremy Lent - On worldviews and meaning-making
10. **Thinking in Systems** by Donella Meadows - On systems dynamics and leverage points

Daniel Schmachtenberger Resources:

- "The Metacrisis" - various talks and essays
- "War on Sensemaking" - on epistemic commons degradation
- "The Wisdom of the Third Attractor" - on alternatives to collapse or control

Organizations and Networks:

- Doughnut Economics Action Lab (DEAL): https://doughnuteconomics.org/
- Democracy in Europe Movement 2025 (DiEM25): https://diem25.org/
- P2P Foundation: https://p2pfoundation.net/
- Next System Project: https://thenextsystem.org/
- Democracy Collaborative: https://democracycollaborative.org/
- Navdanya (Shiva's organization): https://navdanya.org/

Tools and Frameworks:

- Genuine Progress Indicator (GPI) - alternative to GDP
- B Corp Certification - business accountability

- Platform Cooperativism Consortium - democratic digital platforms
- Community Wealth Building - local economic development strategies

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Conclusion: From Empire to Community

The Dynamic Causal Factor Model presented here is not merely a technical framework—it is a blueprint for civilizational transformation. We stand at a crossroads where the "efficiency" of empire has brought us to the brink of self-termination. The Moloch of rivalrous competition compels each actor to choices that collectively doom us all.

The DCFM offers a pathway out: governance that honors complexity, respects boundaries, acknowledges interconnection, and places wisdom above optimization. By mapping decisions through the full web of their impacts—direct, interstitial, and recursive—across all timescales and all affected parties, we can finally see the true cost of extraction and the true value of regeneration.

The choice is ours: We can continue the imperial logic, optimizing for narrow metrics while the foundations crumble. Or we can pivot toward the "Third Attractor"—a world of resilient, regenerative, democratic communities, operating within the Doughnut, honoring the wisdom of diverse knowledge systems, and holding as sacred the trust of future generations.

This framework provides the tools. The implementation requires courage. The outcome determines whether our species' story continues.

As Vandana Shiva reminds us: "The future is not a destination we are going to, but a world we are creating."

Let us create wisely.

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